Limits & Derivatives

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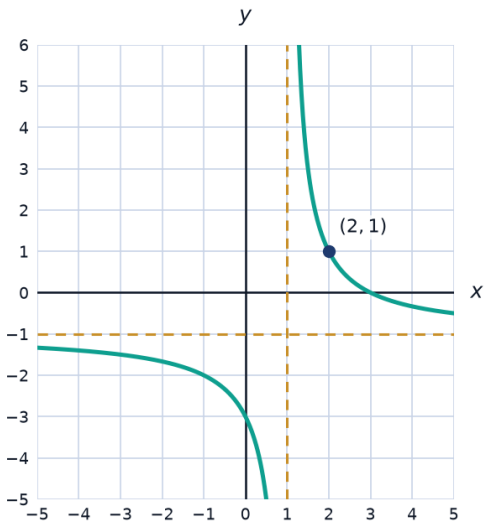
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to 2} (3x^2 - 4x + 1)$.

2 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to -1} (x^3 + 2x - 5)$.



3 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to 0} \frac{5x + 7}{x + 3}$.

4 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to 3} \sqrt{x + 13}$.

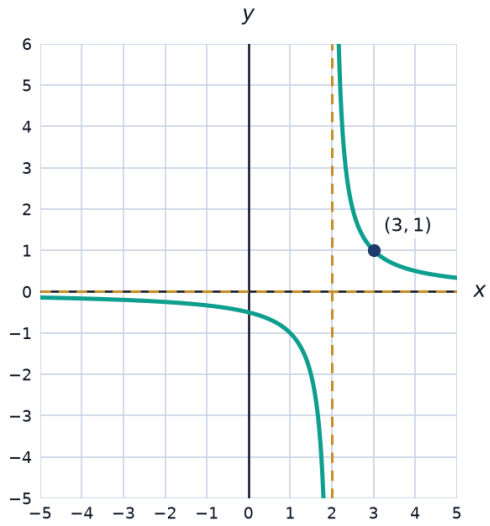
5 **[Foundational]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

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6 [Foundational] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.



7 [Foundational] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x \to 1} \frac{x^2 - 1}{x^2 - 3x + 2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x \to \infty} \frac{3x^2 + 2x}{x^2 - 5}$.

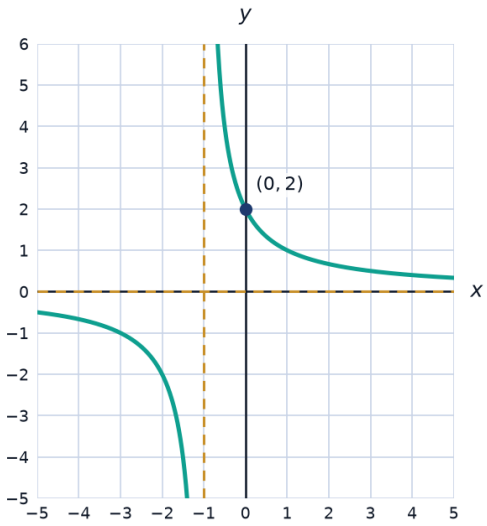
9 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x \to \infty} \frac{2x + 7}{x^2 + 1}$.

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10 **[Intermediate]** • **Target: Limits at infinity**

Evaluate $\lim_{x \to \infty} \frac{5x^3 - 2}{x^3 + 4x}$.



11 **[Intermediate]** • **Target: One-sided limits**

A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x \to 2^-} f(x)$.

12 **[Intermediate]** • **Target: One-sided limits**

Using the same piecewise function, find $\lim_{x \to 2^+} f(x)$.

13 **[Intermediate]** • **Target: One-sided limits**

Using the same piecewise function, does $\lim_{x \to 2} f(x)$ exist? State its value.

14 **[Intermediate]** • **Target: The power rule**

Differentiate $f(x) = x^5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

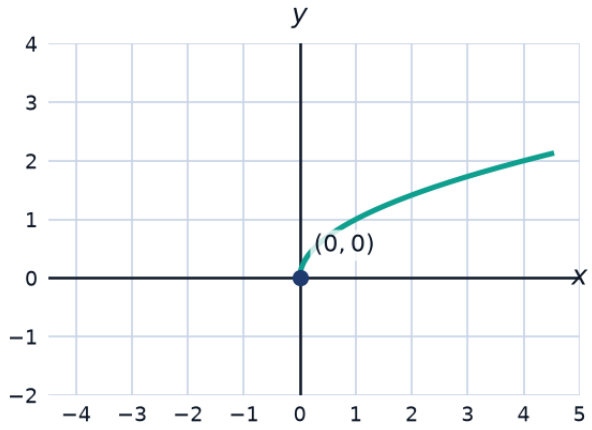
15 **[Challenging]** • **Target: The power rule** • **Cross-Lesson: 4.01 Trigonometry**

Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

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16 [Challenging] • Target: The power rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Differentiate $f(x) = \sqrt{x}$.



17 [Challenging] • Target: The power rule • Cross-Lesson: 7.01 Integration & Applications

Differentiate $f(x) = \frac{1}{x^2}$.

18 [Challenging] • Target: The product rule • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = x^2(3x + 1)$ using the product rule.

19 [Challenging] • Target: The product rule • Cross-Lesson: 9.01 Statistics & Probability

Differentiate $f(x) = (x + 2)(x - 5)$ using the product rule.

20 [Challenging] • Target: The product rule • Cross-Lesson: 1.01 Algebra & Equations

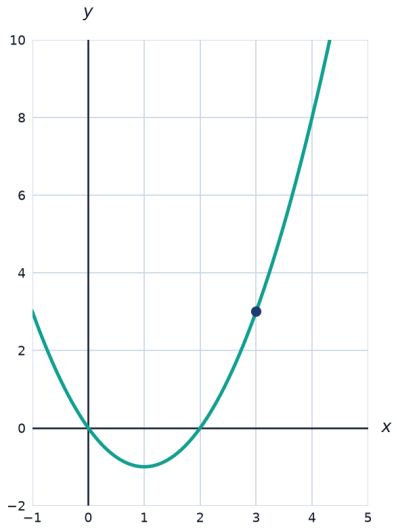
Differentiate $f(x) = (2x - 1)(x^2 + 3)$ using the product rule.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.



a. $y = 3x - 9$

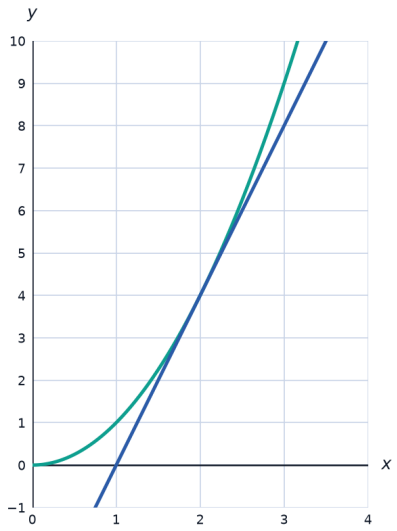
b. $y = 4x - 3$

c. $y = 4x + 9$

d. $y = 4x - 9$

22 [Foundational] • Target: Tangent lines

The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.



a. 4

b. 2

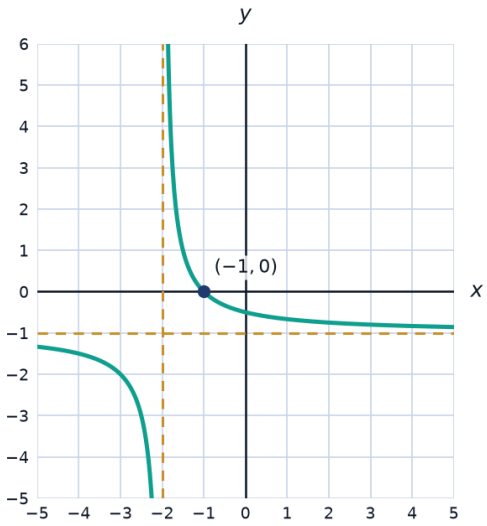
c. 1

d. 8

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23 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to -2} (x^3 - 3x + 4)$.



a. 10

b. -2

c. 2

d. -10

24 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$

25 [Foundational] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x \to -1} \frac{x^2 - 1}{x^2 + 3x + 2}$.

a. -2

b. 2

c. -1

d. $-\frac{1}{2}$

26 [Foundational] • Target: Limits at infinity

Evaluate $\lim_{x \to \infty} \frac{4x^2 + 3x}{2x^2 - 7}$.

a. 2

b. 0

c. $\frac{3}{7}$

d. 4

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27 [Foundational] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{2x^2+1}{5x^3-4}$.

a. ∞

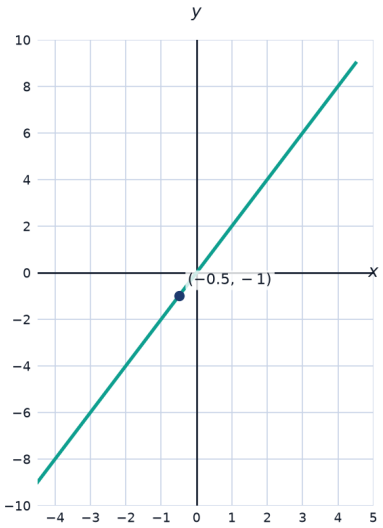
b. $\frac{2}{5}$

c. 0

d. 5

28 [Foundational] • Target: One-sided limits

A function is defined by $f(x) = 2x$ for $x \le 1$, and $f(x) = x + 3$ for $x > 1$. Find $\lim_{x\to 1^-} f(x)$.



a. 1

b. 2

c. 3

d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: One-sided limits

Using the same piecewise function, find $\lim_{x\to 1^+} f(x)$, and state whether $\lim_{x\to 1} f(x)$ exists.

a. 4; does not exist

b. 4; exists

c. 2; exists

d. 2; does not exist

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30 [Intermediate] • Target: The power rule

Differentiate $f(x) = 4x^5 - 3x^3 + 2x$.

a. $4x^4 - 3x^2 + 2$

b. $20x^4 - 3x^2 + 2$

c. $20x^4 - 9x^2$

d. $20x^4 - 9x^2 + 2$

31 [Intermediate] • Target: The power rule

Differentiate $f(x) = 6x^2 - 4x + 9$.

a. $6x - 4$

b. $12x + 9$

c. $12x - 4x$

d. $12x - 4$

32 [Intermediate] • Target: The product rule

Differentiate $f(x) = (x^2 + 1)(x^2 - 1)$ using the product rule.

a. $4x^3$

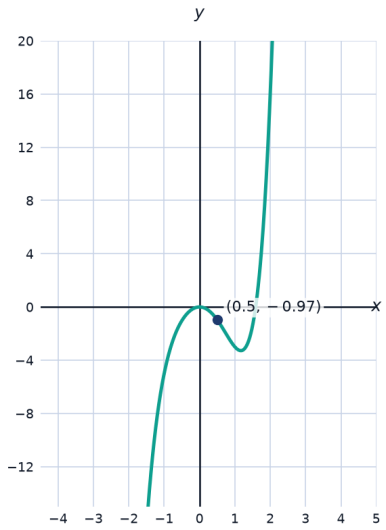
b. $4x$

c. $2x^3$

d. $4x^3 - 2x$

33 [Intermediate] • Target: The product rule

Differentiate $f(x) = x^2(x^3 - 4)$ using the product rule.



a. $5x^4 - 4$

b. $3x^4 - 8x$

c. $5x^4 - 8x$

d. $5x^4 - 8$

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34 [Intermediate] • Target: The chain rule

Differentiate $f(x) = (x^3 + 2)^2$.

a. $6x^2(x^3 + 2)$

b. $6x^2(x^3 + 2)^2$

c. $3x^2(x^3 + 2)$

d. $2(x^3 + 2)$

35 [Intermediate] • Target: Derivatives of trigonometric and exponential functions

Differentiate $f(x) = \cos x - \sin x$.

a. $\sin x + \cos x$

b. $-\sin x - \cos x$

c. $\sin x - \cos x$

d. $-\sin x + \cos x$

36 [Intermediate] • Target: Derivatives of trigonometric and exponential functions

Differentiate $f(x) = 5\cos x$.

a. $5\sin x$

b. $-5\cos x$

c. $5\cos x$

d. $-5\sin x$

37 [Intermediate] • Target: Derivatives of trigonometric and exponential functions

Differentiate $f(x) = e^x - x^3$.

a. $e^x - 3x$

b. $e^x - x^2$

c. $xe^{x-1} - 3x^2$

d. $e^x - 3x^2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Derivatives of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = 2e^x + \sin x$.

a. $2e^x + \cos x$

b. $2e^x + \sin x$

c. $e^x + \cos x$

d. $2e^x - \cos x$

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39 [Challenging] • Target: Increasing, decreasing, and the second derivative • Cross-Lesson: 9.01 Statistics & Probability

Find the second derivative of $f(x) = x^4 + 2x^2$.

a. $12x^2$

b. $4x^3 + 4x$

c. $12x^2 + 4$

d. $12x^2 + 4x$

40 [Challenging] • Target: Word problems: rates of change • Cross-Lesson: 1.01 Algebra & Equations

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the ball's velocity at $t = 2$ seconds.

a. 10

b. -10

c. 30

d. 20

41 [Challenging] • Target: Word problems: rates of change • Cross-Lesson: 2.01 Functions & Transformations

A company's cost to produce x items is $C(x) = 0.02x^2 + 8x + 150$ (in dollars). Find the marginal cost at $x = 100$.

a. 8 dollars

b. 4 dollars

c. 20 dollars

d. 12 dollars

42 [Challenging] • Target: Word problems: rates of change • Cross-Lesson: 3.01 Sequences & Series

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$

b. $t = 5$

c. $t = 6$

d. $t = 2$

43 [Challenging] • Target: Limits & Derivatives • Cross-Lesson: 4.01 Trigonometry

In a laboratory calibration sequence, evaluate $\lim_{x \to -5} (-3x^2 - 5x + 4)$.

a. -49

b. -43

c. 46

d. -46

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44 [Challenging] • Target: Limits & Derivatives • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\lim_{x \to -4} (x^2 - x - 2)$.

a. 18

b. 21

c. -18

d. 15

45 [Challenging] • Target: Limits & Derivatives • Cross-Lesson: 7.01 Integration & Applications

Evaluate $\lim_{x \to 0} (-2x^2 + x - 2)$.

a. 1

b. -2

c. 2

d. -5

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